Sentinel

Predict. Prevent. Intervene.

What is Sentinel??
Real-time AI system that predicts risky situations before they happen by analyzing motion, pose, and audio cues. If prevention fails, it automatically intervenes, assigning tasks and guiding responders to manage the incident. By combining anticipation and autonomous response, it stops accidents before they occur and coordinates help when needed.

What platform are we creating?
Creating a **Web App** + Phone Integration for receiving notifications/messages.

How it works + Inputs?
Very easy to set up an admin panel, takes the camera video feed as input, and it runs through our model. Different sorts of risks. It also prioritizes them, such as water spilled versus someone who has collapsed. Based on that, it identifies the closest AED, for example, or informs the nearest person by sending a notification to their phone with the information they need.

Places to use Sentinel

- Schools
- Warehouses
- Public transit
- Hospitals
- Smart cities
- Prison Cells

Examples of how it works:

1. Two people walking toward a blind corner + motion detection heated movement or arguments. Sentinel predicts a collision and sends subtle alerts to avoid it.

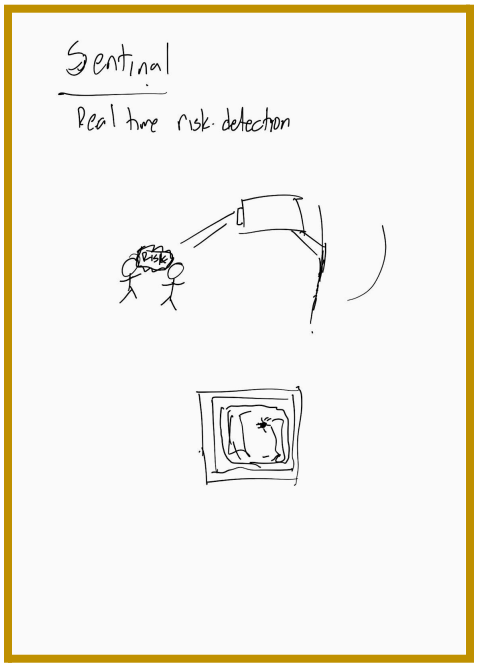
2. Someone falls in a hallway, Sentinel detects the fall and automatically notifies nearby responders with instructions.

3. A heated argument escalates/a fight occurs, Sentinel predicts risk from voice tone and body language, warns participants, and triggers intervention if a conflict occurs.

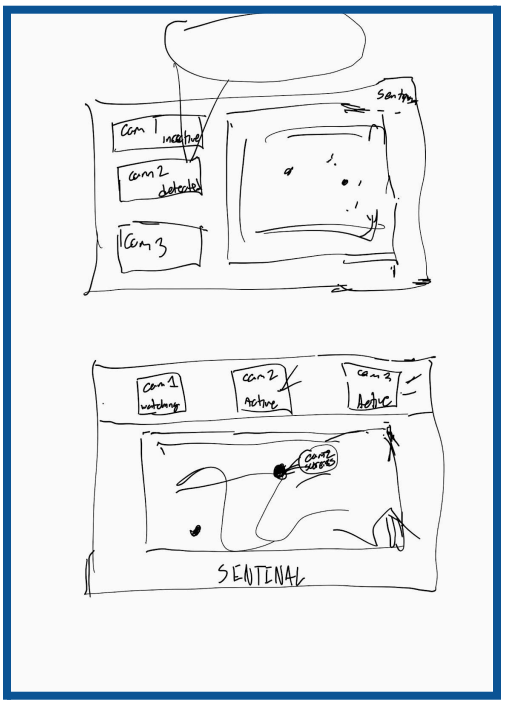
What we want to do in the future:

- Real-time crowd risk mapping
- Predictive aggression modeling
- More complex situations for predicting
- Mobile App integration

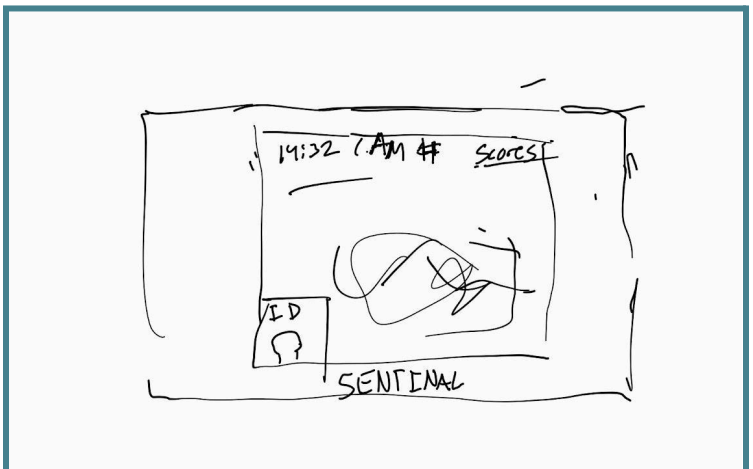
Sketches of Rough Draft:



The camera is capturing video and is run through a model.



Admin panel, how the setup i,s and the camera's which it's capturing and notifications, etc.



Teacher ID's how they receive notifications on

their phone with the message

We need to talk about Privacy that how Sentinel doesn't store the data anywhere

Gemini-3-flash-preview

NLP Model
pose landmarks rather than raw frames whenever possible. The Google MediaPipe Pose

Role-Aware Alert Routing

This is your killer feature.

Different incidents route to different responders.

Example:

Fight risk → security staff
Fall detection → nurse
Crowd panic → administrator

Need to create Mobile APP for the notifications where both Web and Mobile APP Good for notification and like map to the location and like urgency and etc